

Lab 5: HTML Canvas

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Objectives

The objectives of this lab are to:

1. Get more practice with JS dev tools like Node, NPM and Vite.
2. Get more experience using a node package (`tena.js`)
3. Get more practice writing code that uses local storage.
4. Get experience programming with the HTML5 Canvas.

Instructions

Set Up Your Git Repository

1. Create a New GitHub Repository for Lab 5.
Put a `.gitignore` file for node.js in the repository as the first commit.
2. Clone the Repository to your local computer.
3. Download the Starter Files.
Download the starter files for this lab from Moodle and copy them into your local repository folder.
4. Add and commit the new files, then push them up to GitHub.

Study the MemeMaker2 Example

Your instructor has provided a completed web app that demonstrates how to use the HTML Canvas in a JavaScript project. Study this example carefully before building your own app. Here are the things you should look at:

- The Readme.md file—read it! It explains how the app is structured and what it does.
- Run the app, experiment with it and see what it does.
- Look at the source code. For each file read the high-level comments to see what the structure of that module is.
 - Start with the .html page and look to see what it loads and where the `<canvas>` element is.
 - Look at the *MemeUiController.js* file. Note that this is the first module to be loaded and it manages interaction with the DOM (with the exception of canvas API calls).
 - Look at the *memeLogicServices.js* file. Note that this contains a class that does all the logical processing as well as operations that use the canvas API.
 - Look at the modules in the *classes* folder to see how some of the individual features have been implemented in classes.

Complete Your App

Your starter code is fully functional and has no // TODO comments. You will be adding features rather than finishing incomplete code. The features you add depend on the version of the lab you are assigned to do.

Everyone will add these features their lab:

- Automatic storage and retrieval of profiles from local storage.
- Photo filters like *grayscale*, *sepia*, and *contrast* to enhance your image. On your web page, provide filter options (at least 4, of your choice) in a drop-down list.

You can provide using either one of two approaches:

- Use the [lena.js](#) NPM package.
- Use the [canvas filter feature](#) to add filters.
- Something that uses a NPM package. If you added the filters using *lena.js*, then this requirement is fulfilled. If not, you can add another package like [canvas-confetti](#) to put confetti on the page when the user downloads a profile picture.

Version A:

Add these features:

- Circular profile frame, with image or background outside the frame clipped.
- Controls to change background and border color.

- Option to add an outline to the text and choose it's color.
- Drag and zoom your photo to get the perfect crop and positioning.

Version B:

Add these features:

- Controls to change font size and color.
- Option to add an outline to the text and choose it's color.
- Oval profile frame, with image or background outside the frame clipped.
- Stickers (little images) that you can place and drag anywhere on your picture.

Version C:

Add these features:

- Rectangular profile frame with rounded corners and image or background outside the frame clipped.
- Controls to change font family and style.
- Option to add an outline to the text and choose it's color.
- Freestyle drawing that lets you sketch directly on your profile picture with a customizable pen.

Submission

Beta Version and Code Review

- Share your beta version with a lab partner.
 - Your code should be at least 75% complete and working.
 - Share it in your lab partner channel on Discord by posting a [link to your GitHub repository](#).
 - On Moodle, submit the link to your GitHub repository and report:
 - How complete was your web app — was it working?
 - Who did you share your beta code with?
 - What date and time did you share it?
- Commit and push your beta version to GitHub so your lab partner can view the code.
- Do a code review for your lab partner.
 - Post the completed review form on Discord.
 - Submit your code review to Moodle.

- Revise and improve your code based in part on feedback from your lab partner.

Production Version

- Publish to the CIT web server.
 - Upload your completed web app to citstudent.lanecc.edu.
Be sure to upload only the files contained in the `dist` folder (generated by `npm run build`).
 - Test the app on the web server to verify it still works there.
- Submit the following to Moodle:
 - A link to your GitHub repository in the online text.
 - The code review from your lab partner, with the "production" column filled out by you.
 - The citstudent URL for your web app in the online text.

Written by Brian Bird, spring 2026.